ASEAN
GEOAI FUSION 2026

TRAINER / SPEAKER PRESENTATION

GeoAI in Practice: Solving Real-World Telecommunications Challenges

Daniel Marc dela Torre, PhD | Science and Data Lead, Spatial Informatics Group

9 July 2026 • Bootcamp 1 / Pocket Talk • ASEAN GeoAI Fusion 2026

About the Trainer



Daniel Marc dela Torre, PhD

Science and Data Lead, Spatial Informatics Group

Daniel dela Torre, PhD, specializes in remote sensing for environmental monitoring, with expertise in machine learning and time series analysis. He earned his PhD in Geography from the University of Auckland, focusing on climate change impacts on rice yields. Previously, he led the geospatial team in a startup working on the utilities sector.

Geospatial AI

Remote Sensing / Geoinformatics

Environmental and Utilities Monitoring

Session Agenda

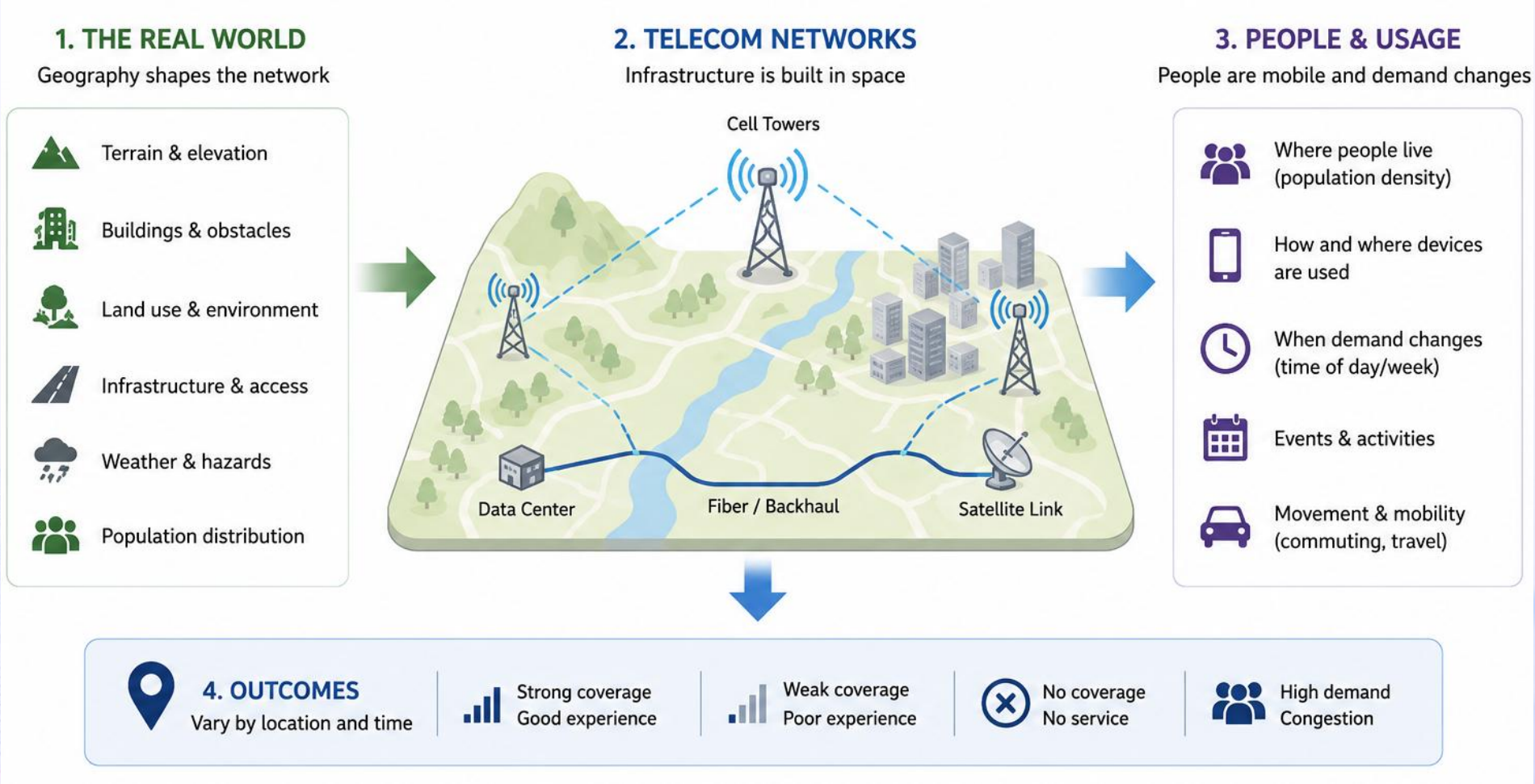
Current Applications of GeoAI in the Telecommunications and Associated Industries

01	**GeoAI in the Context of Telecommunications**	5 min
02	**Challenge 1: Network planning**	5 min
03	**Challenge 2: Demand and service quality**	5 min
04	**Challenge 3: Resilience and sustainability**	5 min
05	**Synthesis and Q&A**	5 min

SECTION 1

GeoAI in the Context of Telecommunications

Geography shapes telecom networks



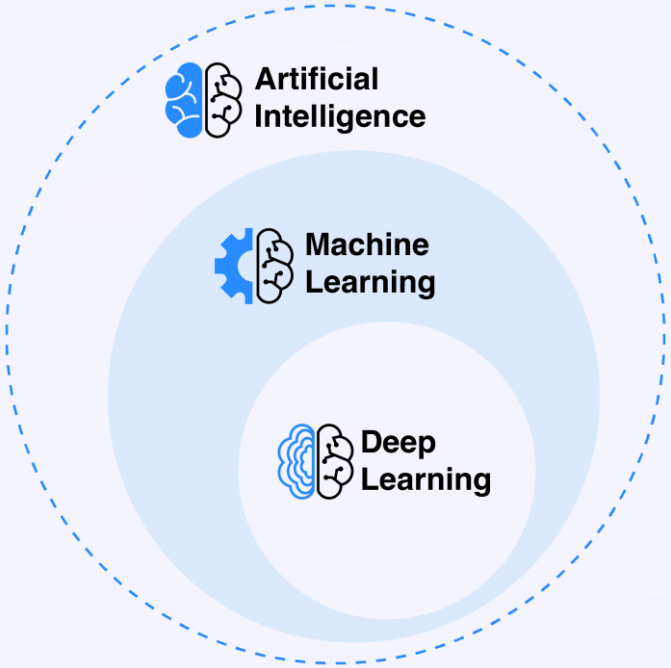
The data paradox

Data Deluge

- Network telemetry
- Customer information
- Market insights
- Inventory

Information Lack

- Up-to-date infrastructure records
- Data quality
- Geographic information
- Resilience/vulnerability

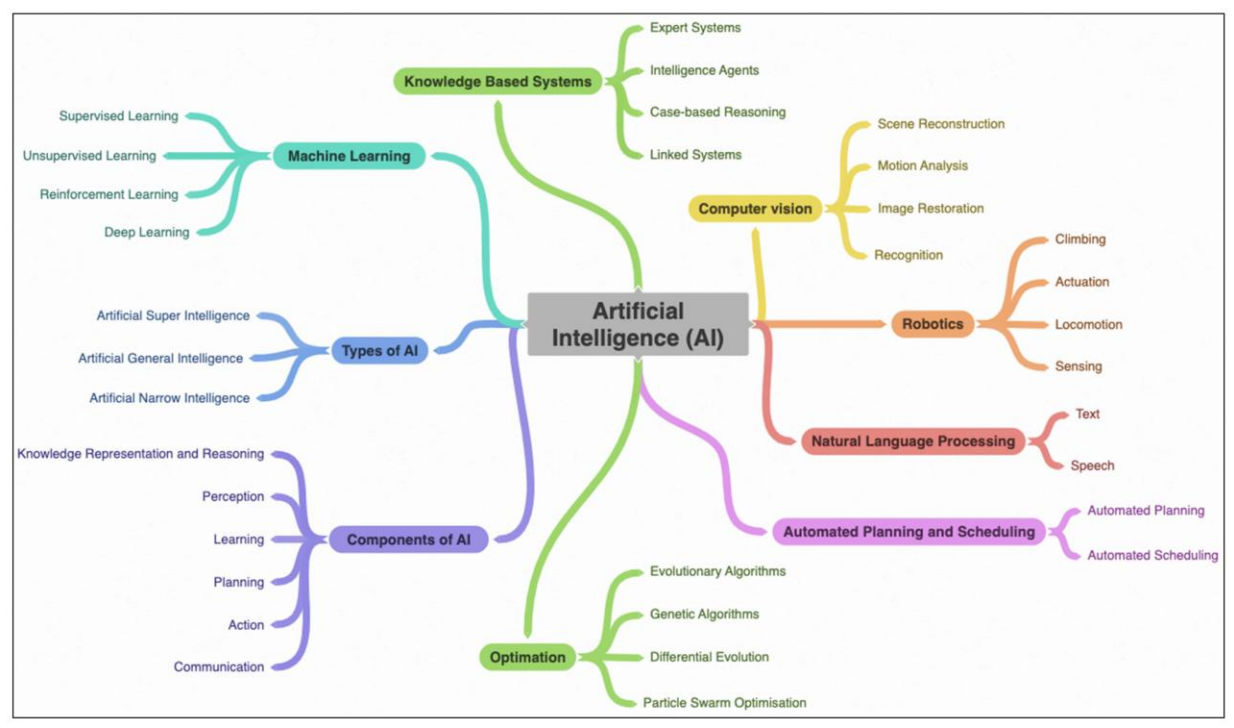


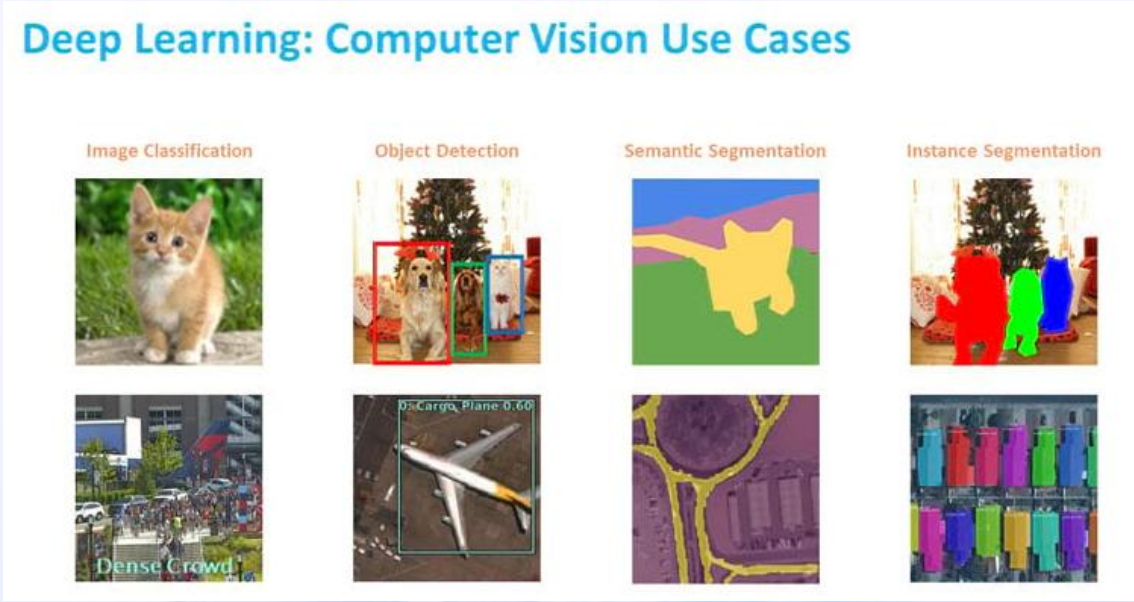
GAP

Actionable Insights

- Where to build?
- What assets are at risk?
- Where are legacy infra?
- How to forecast demand?

Geospatial AI





SECTION 2

Challenge 1: Network planning

Challenge 1: Where Should We Build?

- Telecom planning is a location problem
- Where should a telecom company place the next tower, antenna, fiber route, or 5G small cell?

Why this is difficult:

- Towers are expensive to build.
- Permits and land access can be complicated.
- Signals are affected by buildings, trees, hills, and distance.
- Demand changes by neighborhood, time of day, and type of user.
- 5G and future 6G networks use higher-frequency signals that are more easily blocked.

Geospatial Analysis

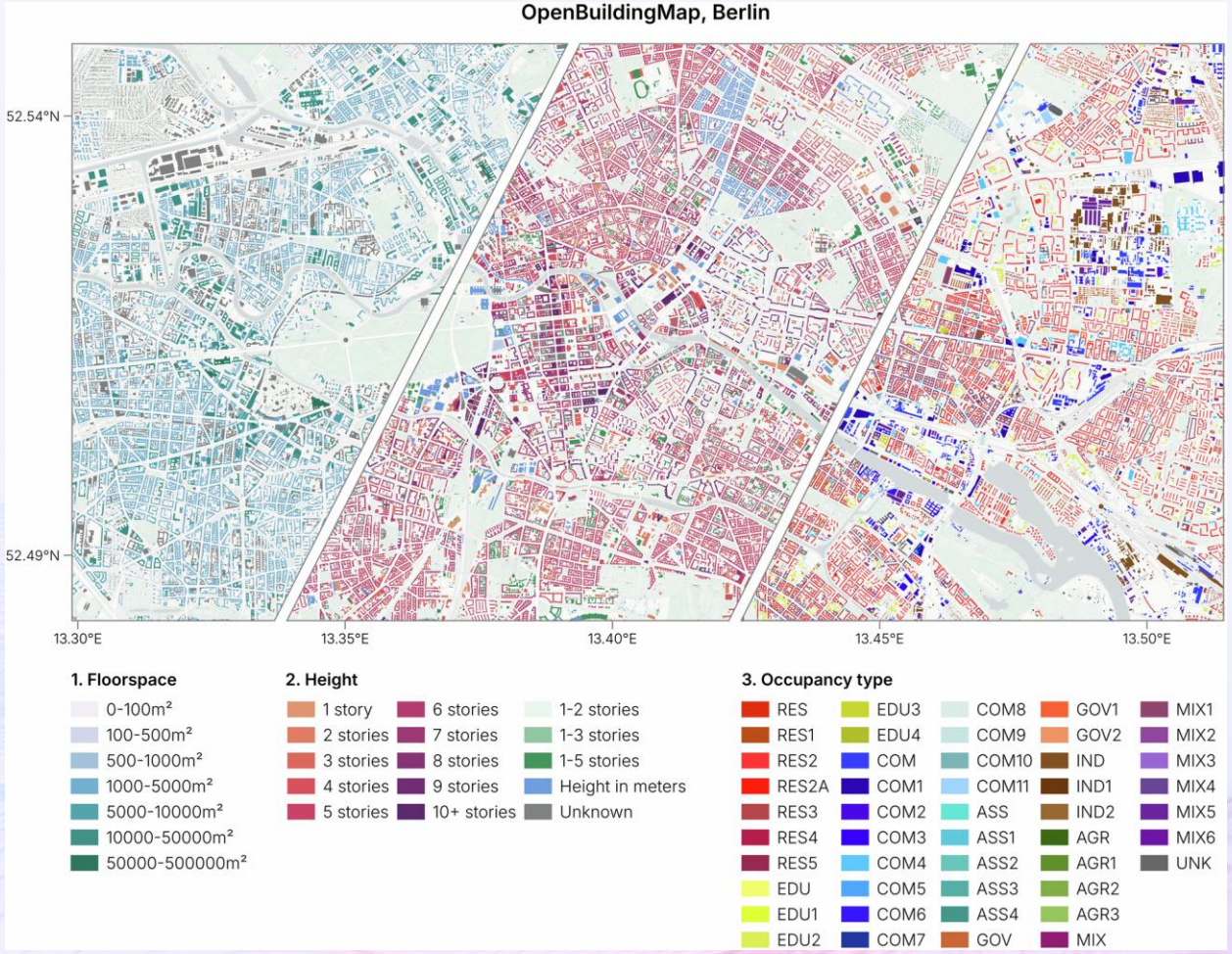
Relevant data layers:

- Population density
- Existing tower locations
- Building height
- Terrain elevation
- Road access
- Fiber routes
- Land ownership
- Customer demand
- Environmental constraints
- Signal coverage maps

Data can be difficult to obtain, update or have quality issues

GeoAI

OpenBuildingMap





- Combines OpenStreetMap, Google Open Buildings, and Microsoft ML Building Footprints, Global Human Settlement Built-up Characteristics

Oostwegel, L.J.N., Schorlemmer, D. & Guéguen, P. From Footprints to Functions: A Comprehensive Global and Semantic Building Footprint Dataset. *Sci Data* **12**, 1699 (2025). https://doi.org/10.1038/s41597-025-06132-z

Automated Point Cloud Processing





- Cleaning, processing, and classifying point clouds used to be a tedious process – AI tools are making this more efficient

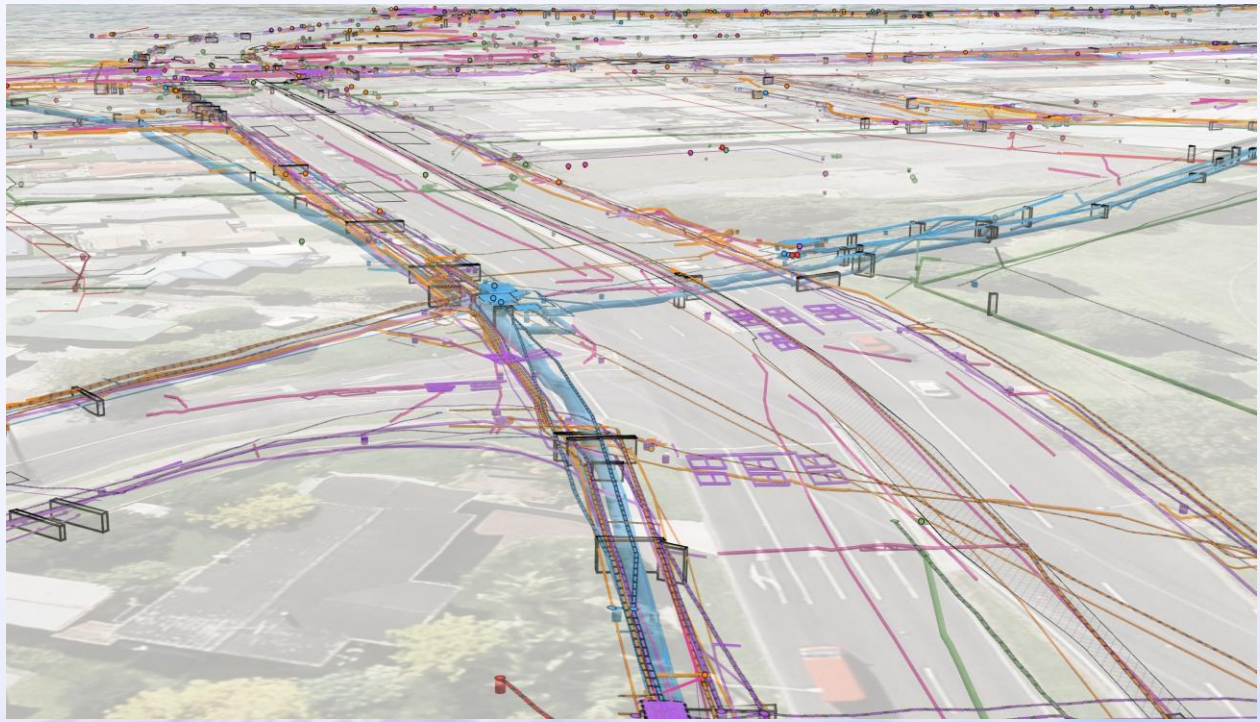
https://www.flai.ai/

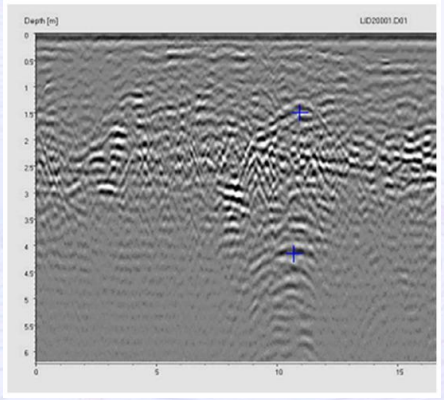
Subsurface Utilities Mapping



PDF / AutoCAD Utilities maps





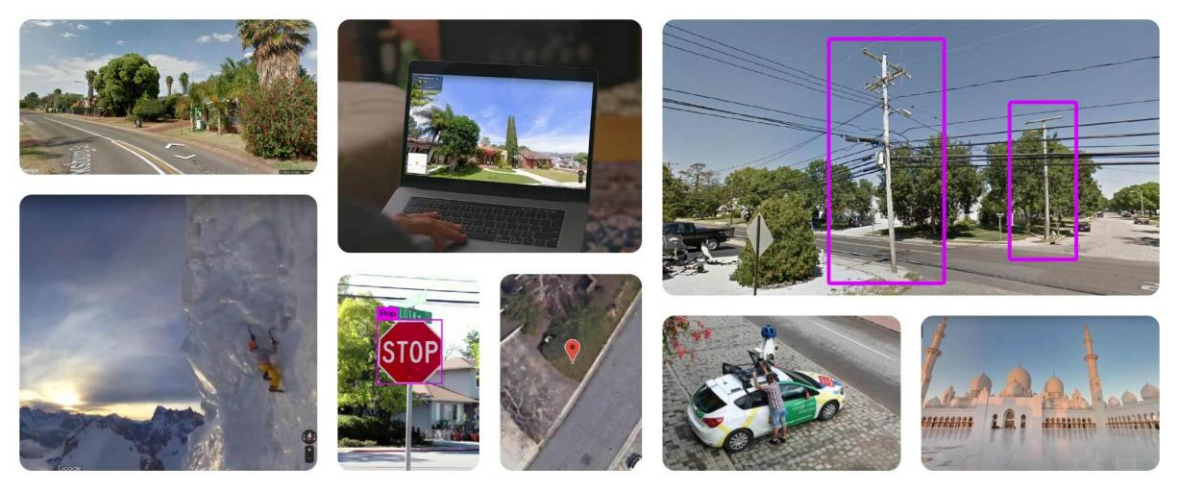


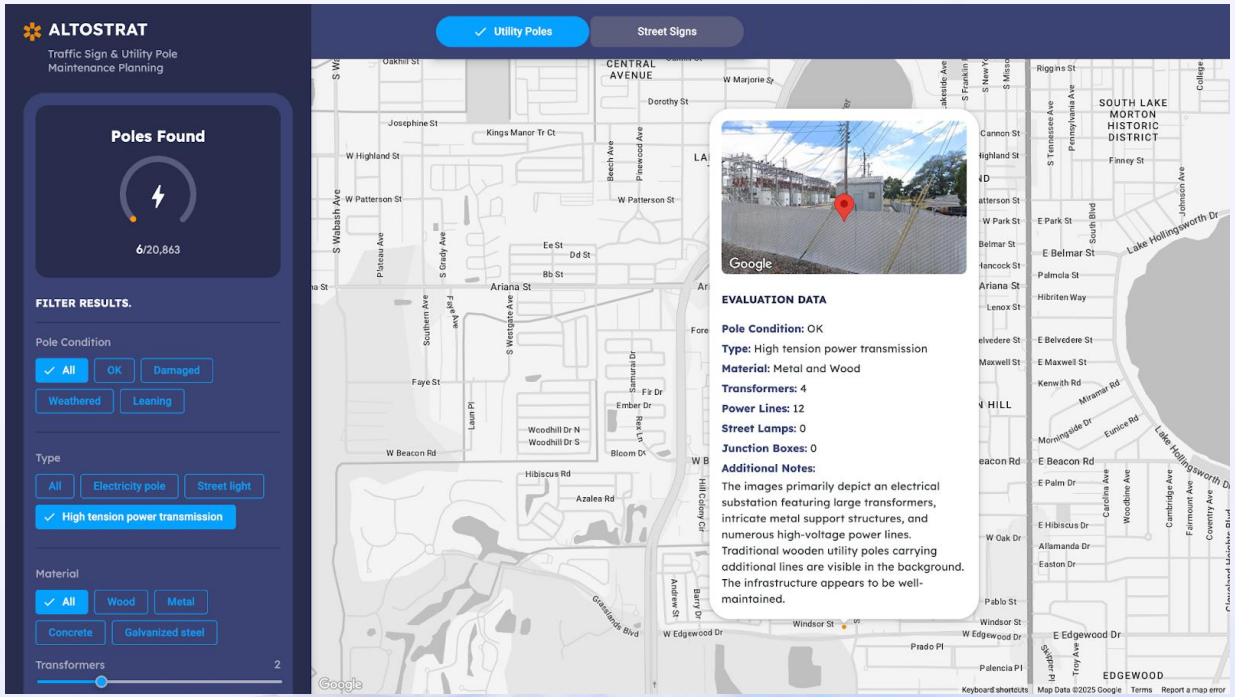


Ground-penetrating Radar

www.undermaps.com
www.reveal.nz
https://www.screeningeagle.com/

Street Imagery for Asset Management





AI is being used with street level imagery to label assets and their condition

https://mapsplatform.google.com/resources/blog/turning-280-billion-images-into-actionable-infrastructure-insights-street-view-insights-is-now-generally-available/

SECTION 3

Challenge 2: Demand and service quality

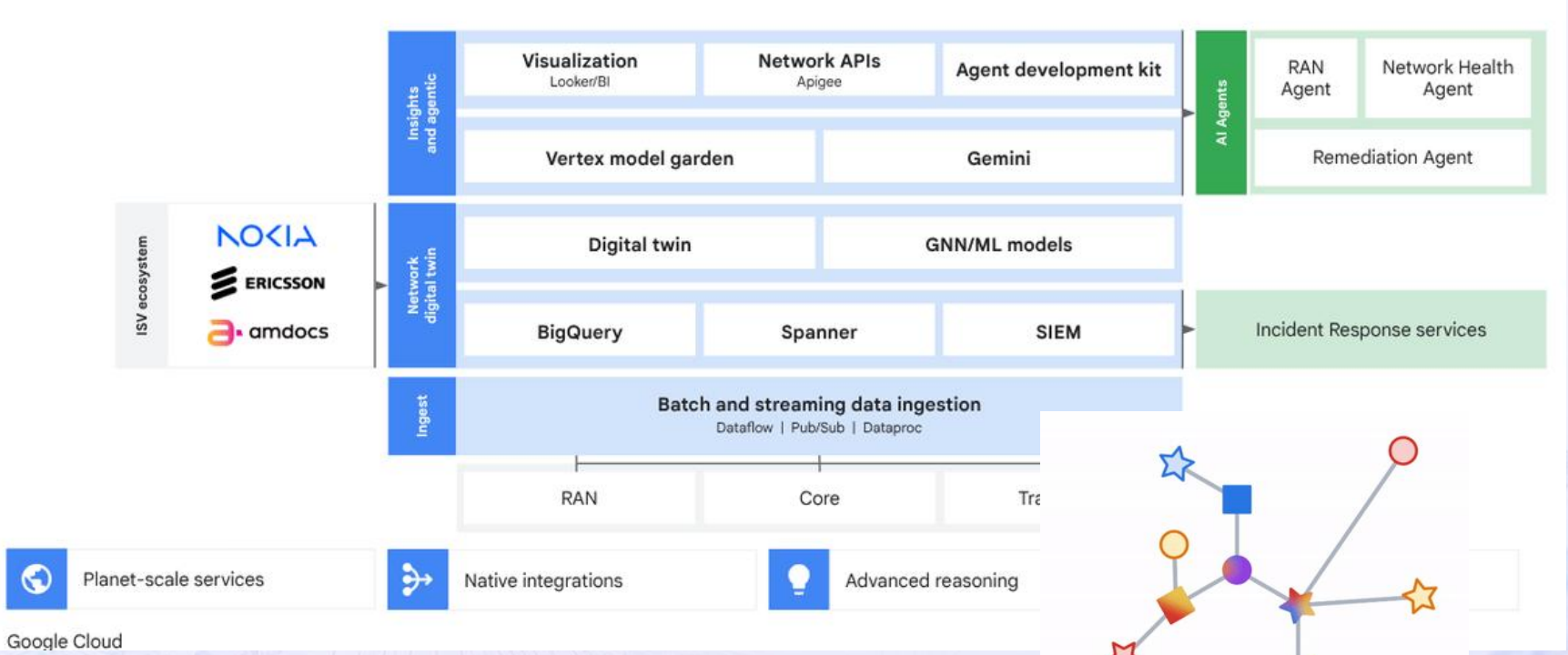
Challenge 2: Why Does the Internet Slow Down?

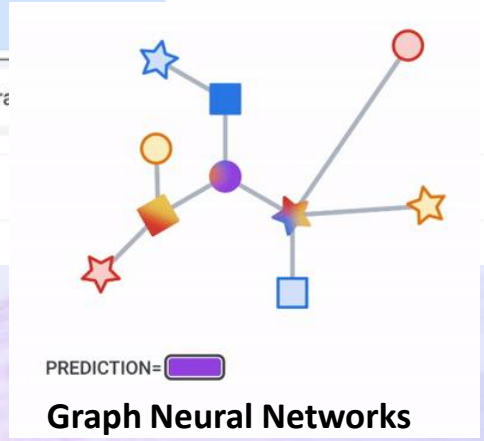
- Network demand changes across space and time

- *Why is mobile service strong in one place but slow in another?*



Autonomous Network Operations





The 3 Strategic Pillars

- Network Digital Twin
- AI Intelligence Layer
- Agentic Framework

Outcomes

- Reduced OpEx
- Faster MTTR
- Predictive Reliability
- Service Agility

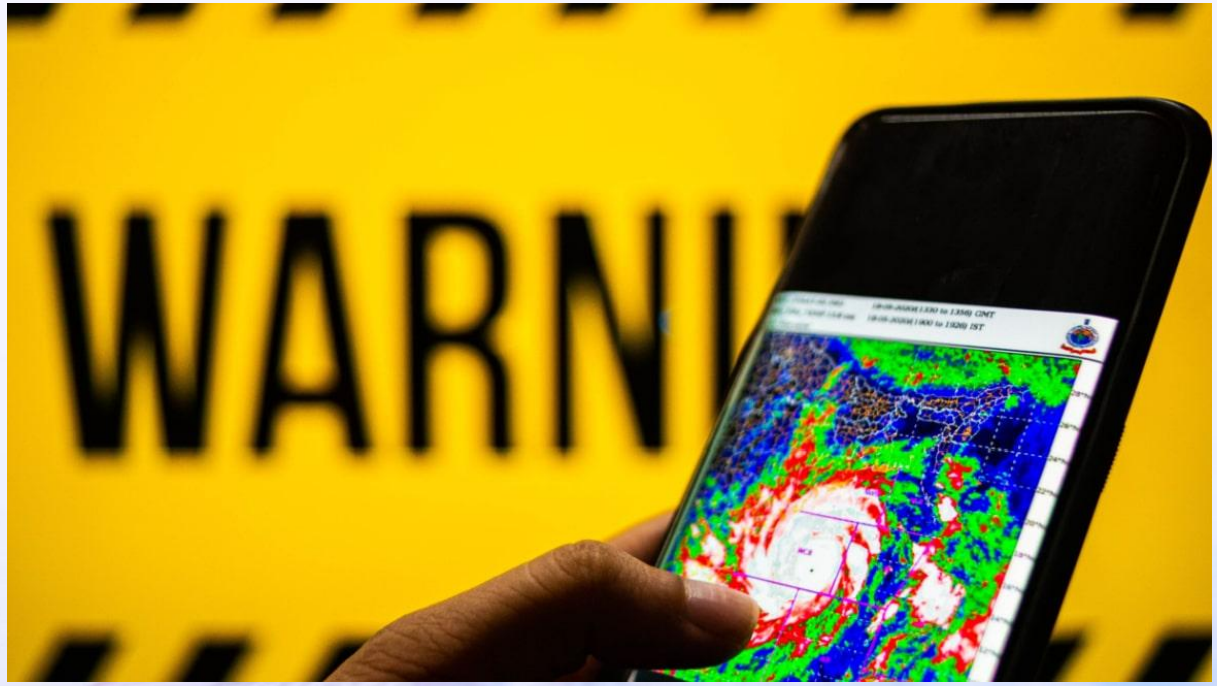
SECTION 4

Challenge 3: Resilience and sustainability

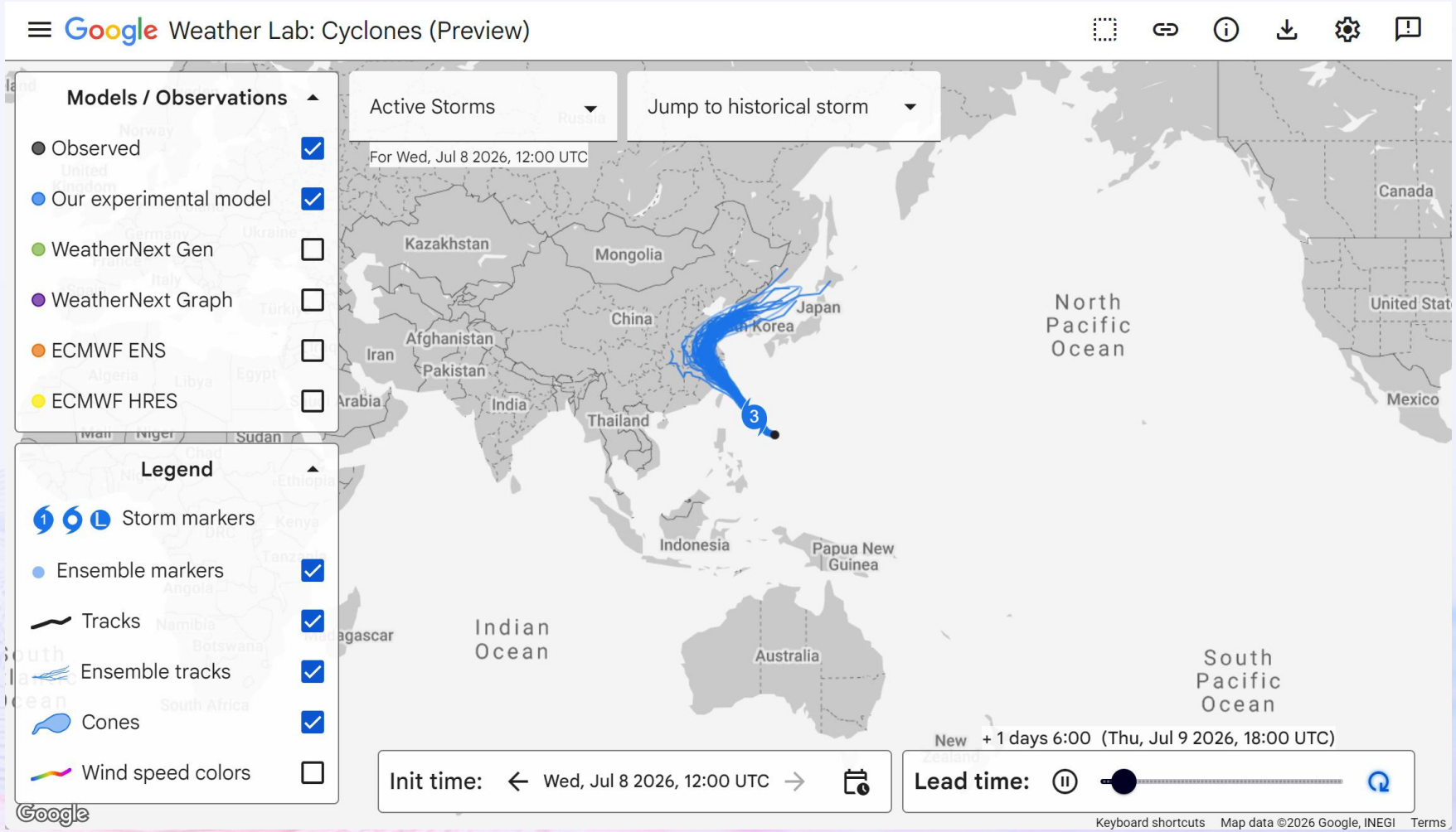
Challenge 3: What Happens When Networks Fail?

- Telecom infrastructure is exposed to real-world hazards

- *How can telecom networks survive floods, storms, wildfires, landslides, heat waves, and power failures?*



Forecasting Typhoons



https://deepmind.google.com/science/weatherlab

ML Forecasting of Floods



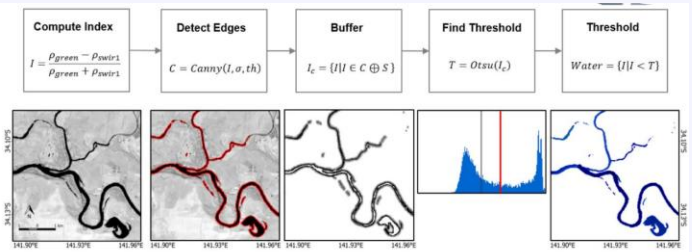
Source: NIWA NZ / Earth Sciences NZ

Multi-Sensor Flood Detection

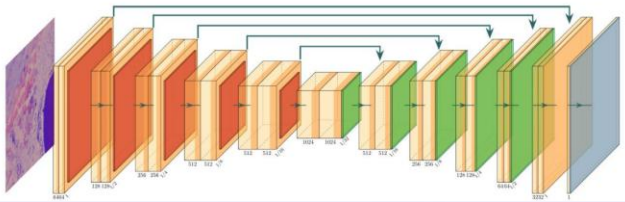
HYDRAFloods

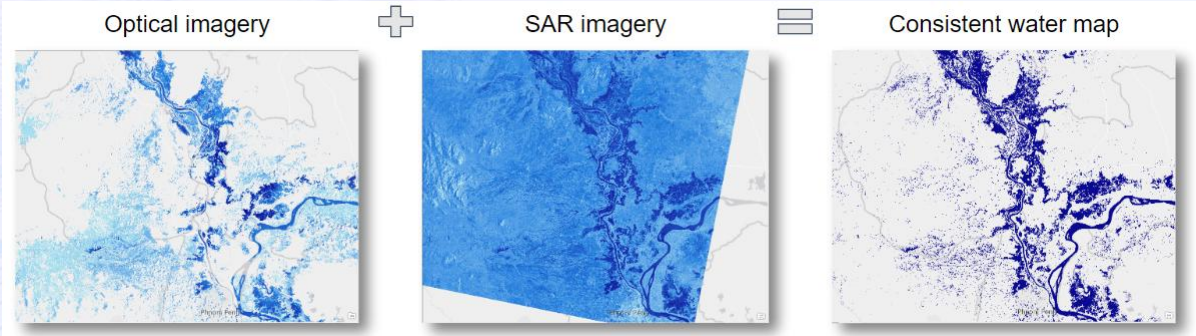
- Single sensor (flood) maps
  - Sentinel-1
  - Sentinel-2
  - Landsat 8

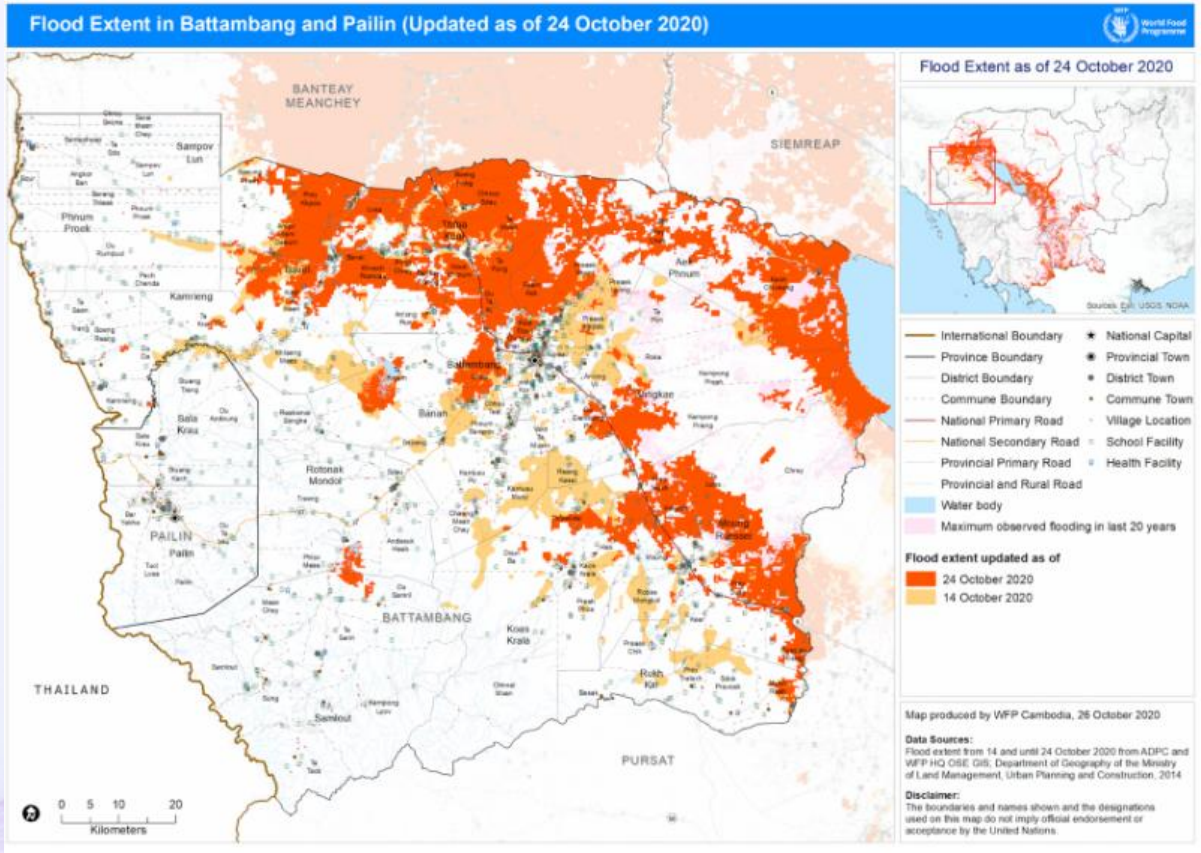
  all → edge-Otsu
  optical → QA UNet
- Daily data fused (flood) maps
  - next slide



Donchyts et al. (2016), Markert et al. (2020)







Source: https://github.com/Servir-Mekong/hydra-floods/tree/master

Damage Assessment



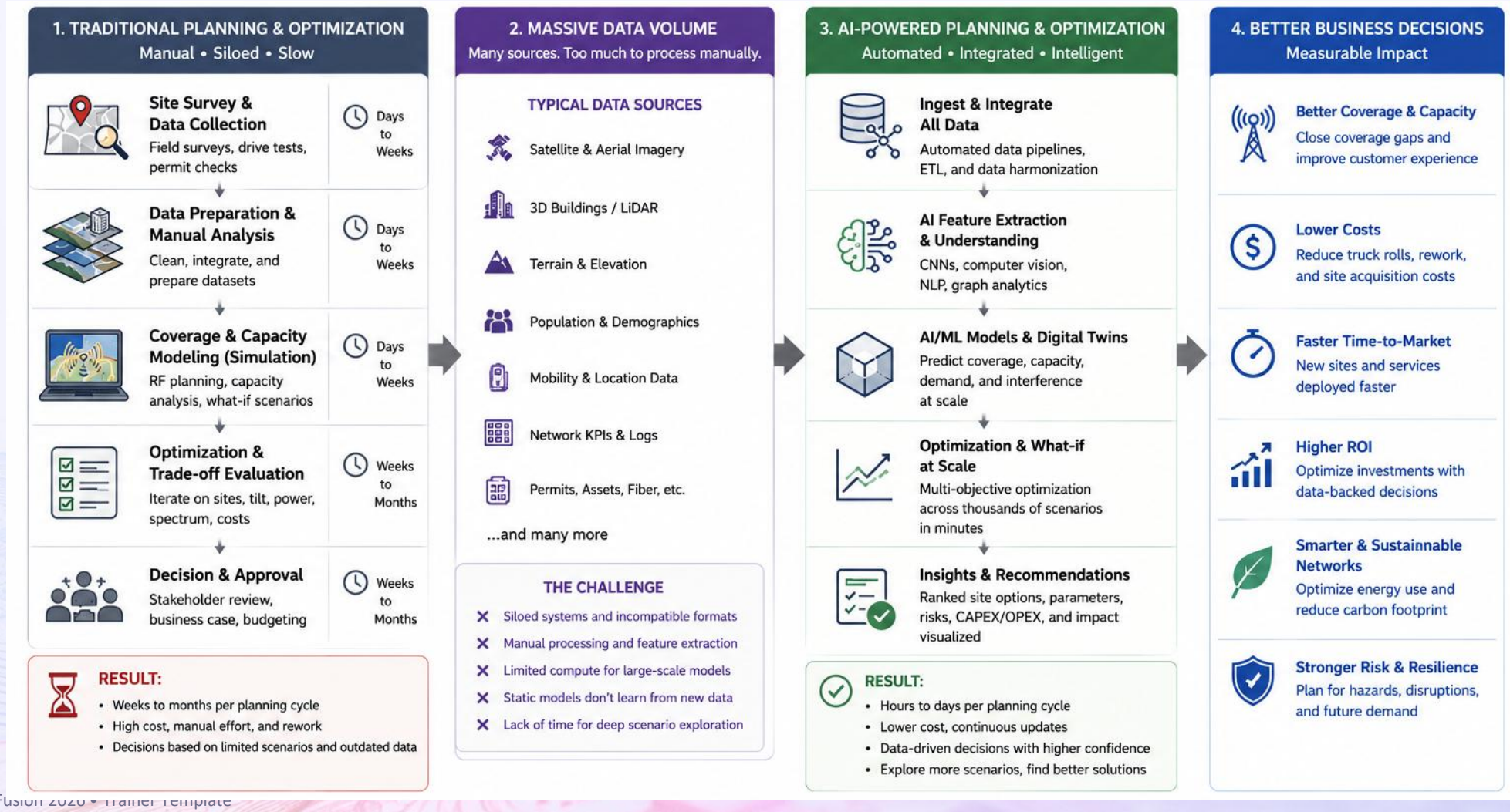


Source: https://storymaps.arcgis.com/stories/0fcd408146334271bc26e5541537d8dd

SECTION 5

Conclusion and Q&A

From Time-Consuming Processes to AI-Powered Insights



Thank You

Questions & Discussion

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